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Asynchrony in the growth and motility responses to environmental changes by individual bacterial cells

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Abstract

Knowing how individual cells respond to environmental changes helps one understand phenotypic diversity in a bacterial cell population, so we simultaneously monitored the growth and motility of isolated motile *Escherichia coli* cells over several generations by using a method called on-chip single-cell cultivation. Starved cells quickly stopped growing but remained motile for several hours before gradually becoming immotile. When nutrients were restored the cells soon resumed their growth and proliferation but remained immotile for up to six generations. A flagella visualization assay suggested that deflagellation underlies the observed loss of motility. This set of results demonstrates that single-cell transgenerational study under well-characterized environmental conditions can provide information that will help us understand distinct functions within individual cells.
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Various physiological states of bacteria are related to their survival strategy under environmental stresses. Normal growth conditions lead a cell culture into a sequence of states generally known as lag phase, exponential phase, stationary phase, and death phase, whereas stressful conditions sometimes induce special states more favorable for survival [1,2]. One example of this phenotypic diversity is a non-growing but metabolically active state called quiescence, in which cells channel available metabolic resources toward particular cellular functions, such as the expression of specific genes [3].

An open question regarding this kind of environment-dependent phenotypic diversity is whether a particular survival mechanism is available to an individual single cell or only to a population of cells because it requires intracellular interactions. This question could be addressed by characterizing what actually happens to individual cells in changing environments and then gradually increase the population size in steps. Although the individuality of bacterial cells in a population has been a topic of interest for decades [4], its direct measurement was faced with the limitations of conventional techniques. As microfabrication techniques have recently become widely available to biologists, however, a variety of sophisticated single-cell cultivation has emerged. Physically confining living cells within small chambers on a microchip is now a promising way to study single cells under rigorously controlled conditions [5]. Microstructures are also used to investigate quorum sensing [6], a way of communication in a bacterial population that depends on the number of cells present in the microenvironment. A step forward in this cultivation concept is to combine it with a cell-handling method so that the number of cells becomes a variable in the environment. We must be able to control the number of cells if we are to evaluate the extent to which a single cell by itself is capable of adapting to its environment.

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We have previously investigated the flexibility of isolated single cells in changing environments and found that the growth and proliferation of single bacterial cells can be suspended and resumed by adjusting the external nutrient conditions [7]. Further characterization of growth-suspended cells would be facilitated by evaluating another phenotypic index parallel to growth and proliferation. The additional index we chose was cell motility driven by the rotation of flagellar motors. Bacterial motility consumes only a few percent of the total available energy [8], which means that tracking how motile the observed cell is would be a sensitive measurement from which to speculate about the metabolic activity of the cell. The simultaneous observation of cell growth and motility might also provide insight into the dynamics of nutrient-governed control of distinct cellular functions within a single cell.

In this paper, we report how the growth and motility of a single cell change in response to a change of the external nutrient condition. Simultaneous monitoring of growth and motility with an on-chip single-cell cultivation system revealed asynchrony in the growth and motility responses to nutrient starvation. Cell growth stopped quickly after starvation began, whereas cell motility was first maintained for several hours and then gradually lost. Discussing these results, we consider energetics within an individual cell, the flagellation state of starved cells, and possible mechanisms responsible for the observed response asynchrony.

Materials and methods

Bacterial strains and culture conditions. An *Escherichia coli* strain AW539 (a Δ*tar* derivative of AW405 [9]) was transformed with a plasmid coding a GFP-tagged aspartate receptor. The resultant strain, AW539/pTar-GFP, exhibited a motility phenotype similar to that of wild-type strains and was used in this study. Its frozen glycerol stock was precultured overnight without shaking in Luria–Bertani (LB) medium (Difco Laboratories; 1% (w/v) tryptone, 0.5% (w/v) yeast extract, 1% (w/v) NaCl; pH 7.0) at 30 °C. For subsequent use the overnight preculture in stationary phase was diluted 1:100 with fresh LB medium.

Microcultivation of isolated single cells. The on-chip cultivation system used in this study is shown schematically in Fig. 1A. As described in detail elsewhere [7,10], it is specialized for the transgenerational observation of isolated single cells under rigorously controlled culture conditions.

Sample cells were collected during their exponential phase, typically 3 h after dilution of the overnight preculture. The collected cells were spread onto an array of SU-8 microchambers created on a biotin-conjugated glass slide (Fig. 1B). The array was then covered with a small piece of avidin-conjugated semipermeable membrane (Spectra/Por Membrane MWCO: 25,000, Spectrum Laboratories), which ensures the confinement of cells while allowing diffusive medium exchange. A medium-circulation module was mounted on the sealed microchambers, which were exposed to flowing culture medium on the stage of an inverted microscope with either a 60× or 100× objective lens. Among the entire array of microchambers, one microchamber containing only one cell was chosen for continuous observation (Fig. 1C). Once initiated, the cultivation was continued under the microscope at 30 °C. The nutrient condition was maintained by the fresh medium flow at the rate of 10 ml/h. Nutrient-rich and nutrient-free conditions were, respectively, LB medium and 0.9% (w/v) NaCl aqueous solution. The observed single cell was always kept isolated from other cells by using optical tweezers (Nd:YAG laser; T20-8S, Spectraphysics) (Fig. 1D): after each cell division, one of the produced sister cells was

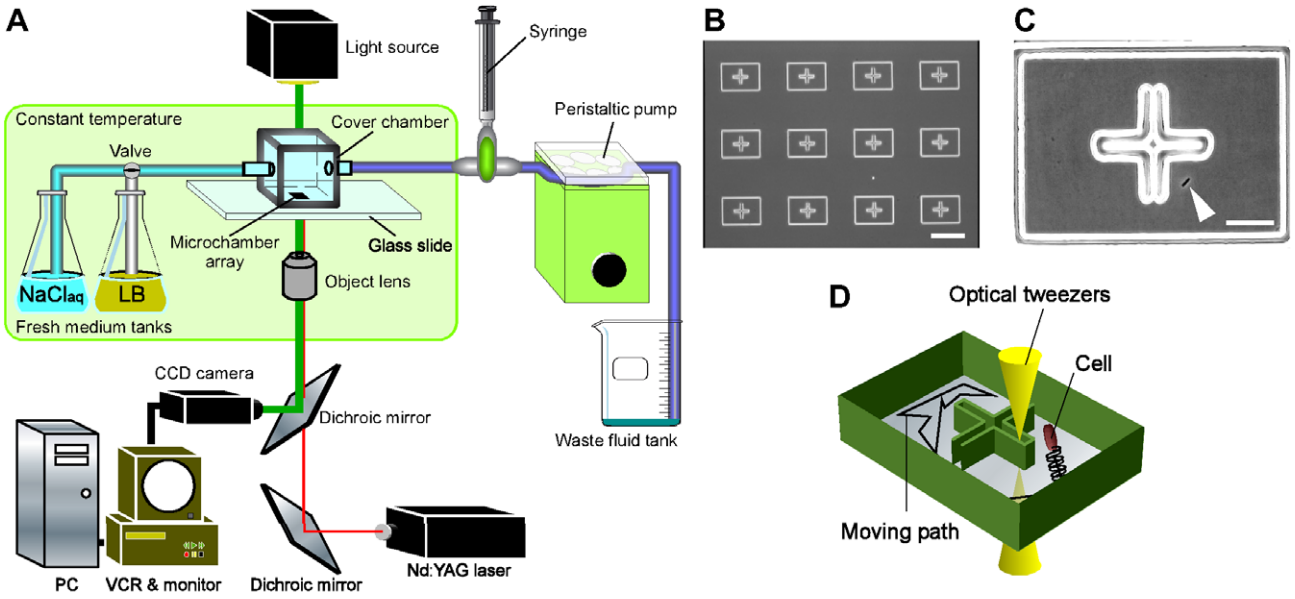


Fig. 1. On-chip cultivation system. (A) System overview. The cells were cultured in microchambers constructed on a glass slide and continuously monitored with a microscope. Constant medium flow defined and maintained the nutrient condition of the observed cells. A focused Nd:YAG laser was used as optical tweezers for non-contact cell manipulation. The entire observation was recorded and analyzed during and after cultivation. A computer with a custom-made program assisted motility analysis. (B) The microchamber array used in this study. Sample cells were spread over the array, and a microchamber containing a single cell was chosen for subsequent observation. (Scale bar, 100 μm.) (C) A typical image recorded during the observation. A phase-contrast image of the swimming single cell (arrowhead) was continuously recorded at video rate (30 frames per second). (Scale bar, 20 μm.) (D) Schematic drawing of the experiment. The observed single cell was allowed to swim inside the microchamber, whereas other cells were carried to the central dead-end region with optical tweezers. The laser irradiation, depriving unwanted cells of their ability to proliferate, was crucial to maintaining the isolated condition over generations.

immediately removed and exterminated by exposure to the Nd:YAG laser at a higher power.

Analysis of single cell growth and motility. All events during continuous monitoring were videotaped for data analysis. To obtain a growth curve, cell length was measured from the recorded image in focus. To obtain a speed curve, the tumbling-free part of cell movement was extracted from the videotape record, and the instantaneous running speed was calculated as the displacement divided by the time required for that displacement. Manual analysis was assisted by a customized computer program that automatically obtained the lateral position of the cell every 0.1 s.

Nutrient condition change by centrifugation. Centrifugation is one of the common methods for exchanging the medium of a batch culture and damages motile bacteria less than dialysis or membrane filtration does [11]. The medium of a population culture was therefore exchanged by centrifugal-washing the culture at 6000*g* for 5 min at room temperature (≈25 °C), diluting it with pre-warmed fresh medium, and then repeating these centrifugation dilution steps until they had been performed ten times. Assuming that each centrifugation-dilution step dilutes the culture medium to 10% of its original concentration, ten steps should have completely removed the nutrient molecules. Again, nutrient-rich and nutrient-free conditions were, respectively, defined as LB medium and 0.9% NaCl aqueous solution.

Flagella visualization. For the fluorescent labeling of bacterial flagella, Alexa Fluor 546 Protein Labeling Kit (A10237; Molecular Probes) was used as reported by Turner et al. [12]. Briefly, the sample cells were washed by centrifugation three times with washing buffer (0.01 M phosphate, 0.067 M NaCl, 10^{-4} M EDTA; pH 7.0), mixed with the kit solution for 1 h, and washed three times with washing buffer containing 10^{-4} % (v/v) Brij 35 surfactant (Sigma). Labeled samples were observed with a microscope equipped with a cooled CCD camera (ORCA II-ER, Hamamatsu Photonics).

Results

Simultaneous observation of single-cell growth and motility under changing environmental conditions

This study demonstrated the feasibility of long-term, transgenerational observation of single-cell growth and motility with the sequential use of multiple medium conditions of choice. We applied nutrient condition changes to an isolated single cell to measure the response of its movement and growth after the environmental changes. In each experiment a motile cell was chosen and then continuously monitored, swimming freely but only within its chamber (Fig. 1D). Transgenerational observation was achieved by using optical tweezers to repeatedly remove sister cells produced by cell division. During the repeated removal the laser power was increased so that any removed unwanted cells were "killed" [13] and permanently fixed to the dead-ends of the chamber. The isolated condition was maintained during medium exchange. The series of above-mentioned procedures provided results presenting the time-course change of cell length and running speed for multiple generations.

Maintenance of motility during starvation

The simultaneous observation of cell growth and motility revealed asynchrony between these two functions when a single cell responded to nutrient starvation (Fig. 2). The cell cycle of the single cell was first checked

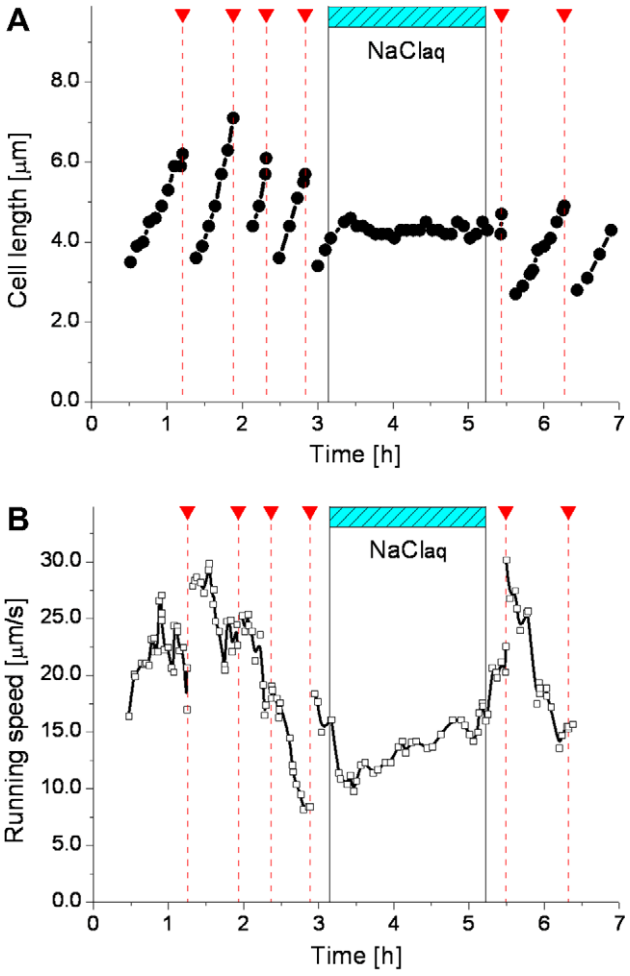


Fig. 2. The effect of 2-h starvation on the single-cell growth and motility. The medium was changed from LB to NaCl solution (hashed) and then back to LB again. (A) The growth curve of a single cell and its direct descendants. (B) The speed curve of the same cells. The triangle symbols with vertical dotted lines indicate the timing of cell division.

in the constant LB (nutrient-rich) medium condition. During each of the first few generations, the growth curve increased monotonically and the speed curve showed a general tendency to decrease. After a few generations, the LB medium was completely exchanged for 0.9% NaCl aqueous solution (nutrient-free) so that the cell started experiencing nutrient starvation. Cell growth stopped abruptly (in 10–20 min), after which cell length (and its shape and volume) remained the same. On the other hand, the cell remained motile with occasional tumbling during starvation. The running speed initially dropped in the first 5 min but then started to increase slightly. After the 2-h starvation period, we exchanged the saline medium for the nutrient-rich LB medium. The cell resumed its growth in about 10 min and then proliferated normally. The running speed increased sharply (in ≈5 min) before the first cell division, and then the cell exhibited the same tendency of decrease in speed that it exhibited before starvation.

Loss of motility after long-term starvation

Whereas the cell maintained its motility during starvation, it lost its motility during an extended starvation period (Fig. 3). When the nutrient-free medium replaced the nutrient-rich medium, the initial response was the same as that seeing during short-term starvation: the cell stopped growing but kept moving, and the running speed dropped within 5 min but then started to increase slightly. A few hours after the medium was exchanged, the running speed reached a local maximum and started decreasing. During this slowing-down process the cell did not swim as smoothly as before, frequently stopping and rotating. Eventually, the cell lost its motility and started to demonstrate Brownian motion (indicated by the arrow in Fig. 3B). Motility was not recovered after the reintroduction of LB medium, but growth and proliferation resumed.

Deflagellation of starved cells in a population

A possible explanation for the loss of motility could be starvation-induced deflagellation (depolymerization of flagellar filaments). To visualize the flagellar filaments, we prepared population samples cultured in LB medium, washed them with nutrient-free NaCl solution by centrifugation, and continued to culture them in the NaCl solution overnight. Although the percentage of motile cells in the samples had decreased from 91% to 12% when the cells had been cultured in the NaCl solution for 20 h, immediately after centrifugation (which took 2 h) it was still 81%. This slow decrease suggests that the loss of motility was caused not by the centrifugation process itself but by nutrient starvation. The swimming behavior of cells remaining motile after overnight starvation was similar to that observed in the single-cell measurement: reduced swimming speed and a rotating motion of the cell body.

The population samples labeled with Alexa 546 showed deflagellation after nutrient starvation. Before starvation, most of the cells cultured in LB were flagellated (Fig. 4A). After overnight (20-h) starvation, on the other hand, almost none of the cells had flagella (Fig. 4B) and those that did had only a single short flagellar filament (Fig. 4C). These cells were no longer motile. GFP expression was confirmed in the labeled cells, which excludes the possibility of contamination during the starvation process.

Discussion

We found that isolated single *E. coli* cells in a nutrient-free condition stopped growing abruptly but remained

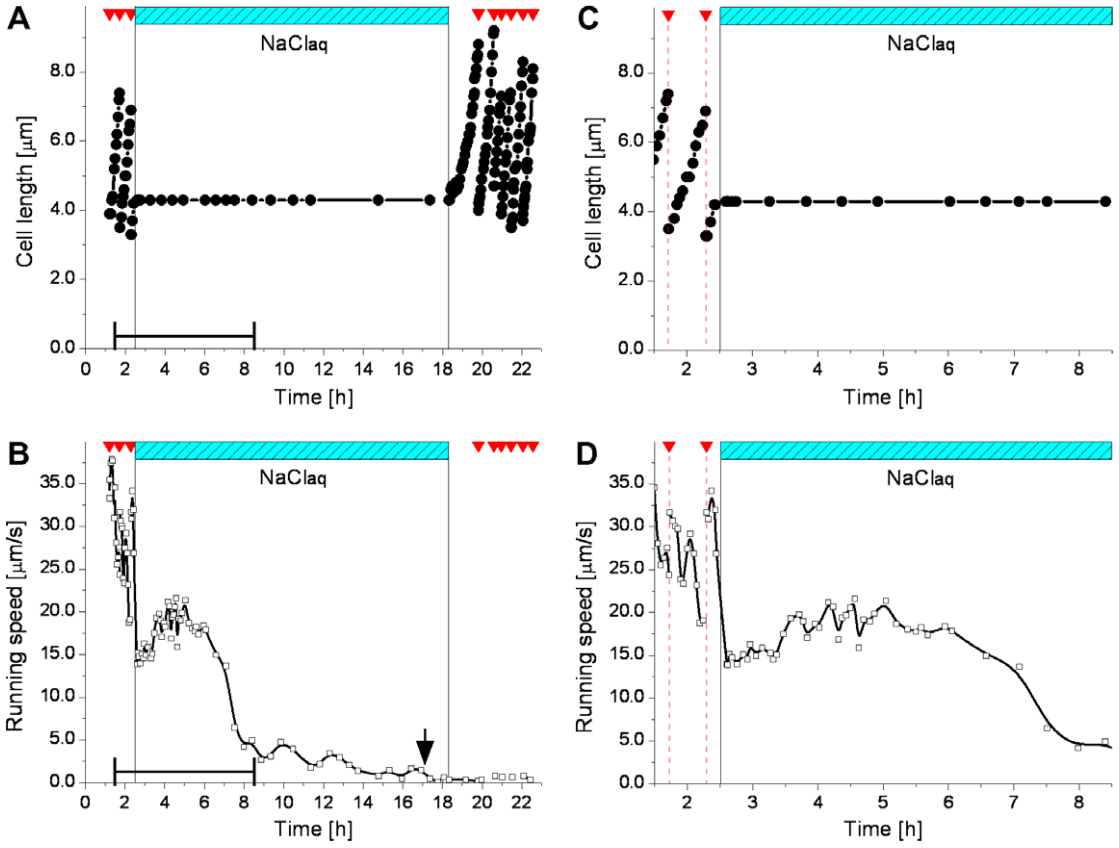


Fig. 3. The effect of 16-h starvation on the single-cell growth and motility. The medium was changed from LB to NaCl solution (hashed) and then back to LB again. (A) The growth curve of a single cell and its direct descendants. (B) The speed curve of the same cells. The triangle symbols indicate the timing of cell division. The arrow in (B) indicates when the cell completely lost its motility and started to show Brownian motion. (C) A magnified graph of (A) showing the region between 1.5 and 8.5 h. (D) A magnified graph of (B) showing the region between 1.5 and 8.5 h.

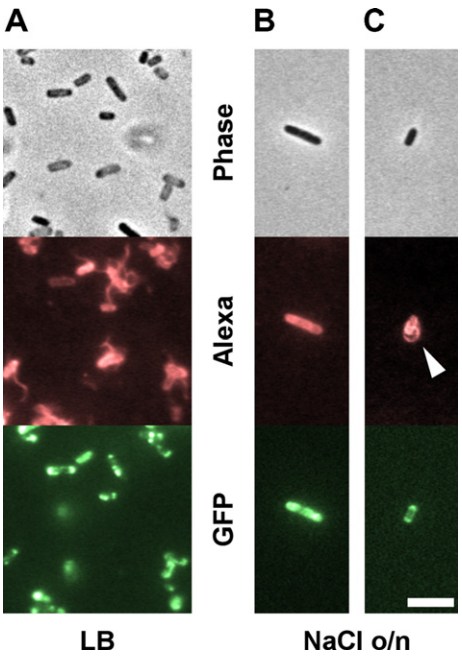


Fig. 4. Flagellation state of starved cells. (A) Control cells cultured in LB medium. Most of the cells had multiple flagella. (B and C) Cells that experienced overnight (20-h) nutrient starvation. Almost all were deflagellated like that shown in (B), but a very few had a short single flagellum (indicated by the arrowhead) like that shown in (C). Phase: phase-contrast images. Alexa: fluorescent images of Alexa 546 that labeled flagella and the entire cell body. GFP: fluorescent images of membrane chemoreceptor protein Tar-GFP for the identification purpose. Note that the fluorescent images are false-colored. Scale bar, 5 μm.

motile for several hours. Since the abrupt stop of growth in response to nutrient deprivation was also observed in our previous study with non-motile, flagella-deficient samples [7], it may be a starvation response common to *E. coli* strains with and without motility. The functional elements responsible for cell growth and cell division seem to be preserved under the starved condition so that growth and division can be readily resumed whenever nutrients are available.

Because growth-suspended cells remain motile, we think that cellular activity gets channeled into selected functions. Like non-growing but metabolically active quiescent cells [3], starved cells might use resources normally allocated to growth and division to maintain the transmembrane proton gradient that drives flagellar rotation. This could be related to a foraging strategy in *E. coli* [14] in that the starving cells would be taking a risk to increase their likelihood of survival by redistributing available energy.

The molecular basis of the maintenance of motility remains to be revealed. Related physiological factors include pH, osmotic pressure, and chemotactic attractants present in the medium. For example, gene regulation is known to be affected by external pH [15]. The effects of these factors can be investigated individually by appropriate combinations of medium conditions.

Motility was maintained for several hours after starvation but was gradually lost. Motility loss was also observed when a bacterial population experienced the centrifugal exchange of medium. Since the starved population consists mostly of deflagellated cells, we think that starvation causes deflagellation (or depolymerization of flagellar filaments). This speculation is consistent with the report that the loss of motility in other adverse conditions (e.g., high temperature or high concentrations of salts) is correlated with a reduction in the amount of cellular flagellin [16] caused by the inhibition of its synthesis [17]. Starvation-induced deflagellation has also been reported in soil bacteria *Rhizobium meliloti* [11].

Isolated single cells that had lost their motility remained immotile for up to six generations (Fig. 3B), but the recovery was confirmed in a batch culture of starved, immotile cells. To test whether on-chip cultivation itself impairs this ability to recover, we did the same on-chip experiment but without cell isolation. In this situation the cell density in a microchamber increased as the starved cells proliferated. After 4 h, when the number of cells had increased to about 50 (corresponding to five to six generations), we saw that a few cells recovered their motility (data not shown). This result indicates that on-chip cultivation itself is not responsible for the inability of isolated single cells to regain their motility.

Whereas the loss of motility is probably caused by deflagellation, the mechanism of motility recovery is not fully elucidated. There are at least three possible explanations that small populations of starved cells seem to be able to recover their motility but isolated single starved cells do not seem to be able to recover their motility. The first is that the probability of this recovery is so low that we did not obtain it in the small number of attempts in the present experiment. The second possible explanation is that the time required to synthesize new flagella from scratch is actually longer than that needed to produce the six generations we observed after starvation (Fig. 3B), even though six generations are enough for the recovery of the normal flagellated state in *Salmonella typhimurium* [18]. The third possible explanation is that starvation causes cells to enter a dormant state in terms of motility and that they can return to their motile state only under conditions requiring the presence of other cells.

What is behind the observed response asynchrony? Or, what is behind the apparent irreversibility in cell motility? Mutations are usually unlikely to alter the genetic information of direct descendant cells, since the per-genome mutation rate in *E. coli* is only 0.0025 per genome replication [19], but the mutation rate in extremely severe environments could be tuned considerably higher as a stress response [20]. The apparent irreversibility might also be due to epigenetic mechanisms that either work within an individual cell or depend on the presence of other cells. Intercellular communication such as quorum sensing [6] or contact-dependent regulation [21] could be related to the mechanisms.

Conclusion

In summary, in this study we investigated the response of isolated single *E. coli* cells to environmental changes.

We focused on the growth and motility of single cells, both of which were monitored continuously and simultaneously. By comparing the time-courses of the changes in these two cellular functions within a single cell, we found asynchrony in the growth and motility responses to nutrient starvation. Cell growth stopped soon after starvation, but cell motility was maintained for several hours under the starved condition. Long-term starvation eventually induced the loss of motility. The recovery of growth also differed from that of motility: although cells quickly resumed growth and proliferation after starvation, they remained immotile for up to six generations. A flagella visualization assay suggested that deflagellation underlies the observed loss of motility. The set of presented results demonstrates that single-cell transgenerational study under strictly defined conditions can provide information that will help us understand distinct functions within individual cells.

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